# University of Cape Coast

**Department of Mathematics & Statistics**

**End of First Semester Examination (2012/2013)**

**MAT 101: Algebra and Trigonometry**

2 HOURS

Answer all questions in Section A, which carries 50% of the total marks, and any two questions from Section B.

## SECTION A

*Answer all questions in this section*
*Four options have been provided for each question. Shade on your answer sheet the letter that bears the same answer as the one you have chosen*

1. Simplify $(P \cap Q') \cup (P \cap Q)$ where $P$ and $Q$ are subsets of the universal set $\mu$.

   **A.** $\phi$ **B.** $Q$ **C.** $P$ **D.** $\mu$

2. Solve $\log_2 2x$

   **A.** $x = \frac{1}{2}$ **B.** $x$
   **C.** $x = 1$ **D.** $x = 2$

3. Solve the absolute value inequality $|4 - x| < 3$.

   **A.** $\phi$ **B.** $\{x : 0 < x < 1\}$
   **C.** $\{x : x < 1 \text{ or } x > 7\}$ **D.** $\{x : x > 7\}$

4. Solve the absolute value inequality $|x + 3| > 4$.

   **A.** $\{x : x < -7 \text{ or } x > 1\}$
   **B.** $\{x : x < -1 \text{ or } x > 1\}$
   **C.** $\{x : x > 1\}$
   **D.** $\{x : x < -7\}$

The polynomial defined as $p(x) = x^3 + a_2x^2 + a_1x + a_0$ has zeros $r_1$, $r_2$ and $r_3$.

Use this information to answer Questions 5 to 7.

5. Find a relation between the sum of the zeros and the coefficients of $p(x)$.

   **A.** $r_1 + r_2 + r_3 = a_1$ **B.** $r_1 + r_2 + r_3 = -a_2$
   **C.** $r_1 + r_2 + r_3 = -a_0$ **D.** $r_1 + r_2 + r_3 = a_2$

6. Which of the following relationship is correct?

   **A.** $r_1 + r_1r_2 + r_1r_3 = a_1$ **B.** $r_2 + r_1r_2 + r_2r_3 = a_1$
   **C.** $r_3r_2 + r_1r_3 + r_2r_3 = a_1$ **D.** $r_1r_2r_3 = a_0$

7. If $p(x) = x^3 + 3x^2 - x - 3$ and $-1$ is a zero of $p(x)$, find the other zeros.

   **A.** $(1, 3)$ **B.** $(3, -1)$ **C.** $(-3, -1)$ **D.** $(-3, 1)$

8. Solve the inequality $-3 \le 2x + 1 \le 5$.

   **A.** $-1 \le x \le 1$ **B.** $-1 \le x \le 3$
   **C.** $-2 \le x \le 2$ **D.** $-2 \le x \le 3$

A function $f(x)$ is given by $f(x) = \frac{x}{x^2 - 1}$.

Use it to answer Questions 9 and 10.

9. Which of the following is a vertical asymptote?

   **A.** $x + 1 = 0$ **B.** $x = 0$
   **C.** $2x + 1 = 0$ **D.** $x - 2 = 0$

10. Find the horizontal asymptote of $f(x)$.

    **A.** $x$-axis **B.** $y = -1$ **C.** $y = 1$ **D.** $y = 2$

11. Given $P(x) = (x - a)^3$, the following are all true except that

    **A.** $x = a$ has multiplicity 1
    **B.** $(x - a)^2$ is a factor of $P(x)$
    **C.** $P(x)$ is of degree 3
    **D.** $(x - a)$ is a factor of $P(x)$

12. Under what condition is a function $f(x)$ considered to be odd?

    **A.** $f(-x) = f(x)$ **B.** $kf(x) = f(kx)$
    **C.** $f(-x) = -f(x)$ **D.** $f(x) = k$

13. Find the coefficient of the term in $x^{\cdot}$ in $\cdot$

A. 220 B. 495 C. 1760 D. 1080

14. All the following are characteristics of $\tan x$ except that it is

A. an odd function
B. symmetrical about the origin
C. periodic with period $2\pi$
D. asymptotic at $x = n\pi +$ $\cdot$

15. One of the properties of $\sec x$ is that

A. it is symmetrical about the y-axis
B. it is an odd function
C. its axes of symmetry are $x = \pi$
D. it is periodic with period $\pi$

16. What is the total number of terms in the expansion of $\left[(a+b)^3\right]^{17}$.

A. 52 B. 51 C. 50 D. 20

17. Let $A$ be $n \times m$ matrix and $B$ a $n \times k$. What is the dimension of $A(A^T B)$.

A. $n \times m$ B. $m \times m$ C. $n \times k$ D. $k \times m$

18. Find $x$ if $\cos(x - 30^\circ) = 0.5$.

A. $60^\circ$ B. $240^\circ$ C. $300^\circ$ D. $330^\circ$

19. Find the determinant of $\mathbf{A} = \begin{pmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{pmatrix}$.

A. 240 B. 400 C. 480 D. 560



20. Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$ be a polynomial, where $a \in Z^+$. If $\frac{p}{q}$ is a zero of $P(x)$, which of the following is *not* true?

A. $p$ is a factor of $a_0$ B. $p$ is a factor of $a_1$
C. $q$ is a factor of $a_n$ D. $\frac{p}{q}$ is rational

21. Find the exact value of $(1.02)^3$.

A. 0.02 B. 0.014 C. 1.1041 D. 0.0014

22. If **M** and **N** are non-singular matrices, then for **MN**

A. $|MN| \neq 0$ B. its inverse does not exist
C. its transpose does not exist D. $|MN| = 0$

A binomial expression is given as $(x+2)^{10}$.

Use it to answer Questions **23** and **24**.

23. Find the 5th term of the expansion of the expression in descending powers of $x$.

A. $5040x^5$ B. $360x^6$ C. $3360x^6$ D. $13440x^4$

24. Find the 5th term of the expansion of the expression in ascending powers of $x$.

A. $3360x^6$ B. $13440x^4$ C. $360x^4$ D. $130x^6$

25. Find the truth set of $\log_{10}(2x^2 + 5x + 97) = 2$.

A. $\{-\frac{1}{3}, -3\}$ B. $\{\frac{1}{3}, 3\}$ C. $\{-\frac{1}{2}, -1\}$ D. $\{\frac{1}{2}, -3\}$

26. For what value of $k$ does the equation $4x^2 - 8x + k = 0$ have equal roots?

A. $k = 0$ B. $k = 2$ C. $k = 4$ D. $k = 8$

A quadratic function $f(x)$ has zeros $m+\frac{1}{n}$ and $n+\frac{1}{m}$ where $m$ and $n$ are roots of the equation $px^2+qx+p=0$

Use it to answer Questions 27 to 29

27. Find the sum of the roots

A. $\frac{p(q+1)}{q}$

B. $\frac{(q+1)}{p}$

C. $\frac{-q(p+1)}{p}$

D. $\frac{p(q-1)}{q}$

28. Find the product of the roots

A. $\frac{-q(p+1)}{p}$

B. $\frac{(q^2-2q+1)}{p}$

C. $\frac{(q^2-2q-1)}{q}$

D. $\frac{(q^2+2q+1)}{q}$

29. Find the equation whose roots are $m+\frac{1}{n}$ and $n+\frac{1}{m}$.

A. $x^2+\frac{(q+1)}{p}x+\frac{(q^2-2q-1)}{q}=0$

B. $x^2+\frac{p(q-1)}{q}x+\frac{(q^2-2q+1)}{p}=0$

C. $x^2+\frac{xp(q+1)}{q}+\frac{(q^2+2q+1)}{q}=0$

D. $x^2+\frac{p(q-1)}{q}+\frac{(q^2+2q+1)}{q}=0$

30. If $\theta$ is obtuse, and $\sin\theta=\frac{4}{5}$, find $\cos\theta$.

A. $-\frac{3}{4}$ B. $-\frac{3}{5}$ C. $\frac{3}{5}$ D. $\frac{3}{4}$



## SECTION B

*Answer two questions only from this section*
*All questions carry equal marks*

1. (a) If $\alpha$ and $\beta$ are the roots of $2x^2-x-2=0$, without solving the equation.

(i) Find the value of $\alpha^2-\beta^2$

(ii) Show that $\alpha^3-\beta^3=\frac{5}{2}(\alpha^2-\beta^2)$

(iii) Find the equation whose roots are $\alpha^3$ and $\beta^3$

(b) Factorize completely the polynomial

$$P(x)=6x^3+13x^2-19x-12$$

2. (a) Use matrix methods to solve the system of equations

$$\begin{aligned}3z-y+2x+17&=0\\2z+6y-10&=0\\-z-2y+x+3&=0\end{aligned}$$

(b) Given the matrices

$$A=\begin{pmatrix}3&2&-2\\1&-3&1\\2&1&-3\end{pmatrix} \text{ and } B=\begin{pmatrix}8&4&-4\\5&-5&-5\\7&1&-11\end{pmatrix},$$

find $AB$ and deduce the inverse of $A$.

(c) If $X=\begin{pmatrix}x&1\end{pmatrix}$, $x\in\Re$, $P=\begin{pmatrix}1&-2\\-2&3\end{pmatrix}$ and $X^t$ is the transpose of $X$, find the truth set of $(XP)X^t=0$.

3. (a) Solve the equation

$$\sec\theta\tan\theta = 2,$$

giving solution for $0^\circ < \theta < 360^\circ$.

(b) Prove that $\tan 3\theta = \dfrac{3t - t^3}{1 - 3t^2}$, where $t = \tan\theta$.

Hence, or otherwise, show that

$$\tan\frac{\pi}{12} = 2 - \sqrt{3}.$$

4. (a) Sketch and describe completely the curve

$$f : x \mapsto 2 - \sqrt{3 - 2x - x^2}$$

(b) The sequence

$$3 - \sqrt{2},\ 11 - 6\sqrt{2},\ \ldots$$

is a geometric progression. Find the common ratio and the sixth term.